

Options

Title: Not titled yet

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Output Language: Python

Generate Options: QT GUI

Variable

ID: samp_rate

Value: 1M

Variable

ID: sps

Value: 6

Constellation Object

ID: constell0

Constellation Type: QPSK

Soft Decisions Precision: 8

QT GUI Range

ID: frequency

Default Value: 2.1G

Start: 500M

Stop: 2.5G

Step: 100M

Import

Import: math

Variable

ID: alpha

Value: 200m

sample_per_symbol

alpha = 0.2

bandwidth -

The transmitter block diagram illustrates the signal processing flow. It begins with a **Random Source** (Minimum: 0, Maximum: 255, Num Samples: 50M, Repeat: No) providing random bytes. These bytes are processed by an **Unpack K Bits** block (K: 8, byte -> bit). The output is split: one path goes to a **Virtual Sink** (Stream ID: ber_input_signal), and the other goes to a **Pack K Bits** block (K: 2, bit -> 2bit (complex)). The packed bits then enter a **Constellation Modulator** (Constellation: <gnu...f0ef70>, Differential Encoding: Yes, Samples/Symbol: 6, Excess BW: 200m). The modulator's output is split again: one path goes to a **Constellation Encoder** (Constellation Object: ...f70>) which outputs to a **Virtual Sink** (Stream ID: bits_tx), and the other path goes to a **Channel Model** (Noise Voltage: 0, Frequency Offset: 0, Epsilon: 1, Taps: 1, Seed: 0, Block Tag Propagation: No). The channel model's output enters a **Root Raised Cosine Filter** (Decimation: 1, Gain: 1, Sample Rate: 1M, Symbol Rate: 250k, Alpha: 200m, Num Taps: 66). The filter's output is then processed by a **Throttle** block (Sample Rate: 8M, Limit: None) before being sent to a **Virtual Sink** (Stream ID: rx_signal_filtered). This virtual sink is connected to two external sinks: **Soapy BladeRF Sink** and **Soapy PLUTO Sink**, both configured with a Sample Rate of 1M and Center Freq (Hz) of 2.1G.

The receiver block diagram shows the signal processing flow from external sources to various sinks. It starts with two **Soapy** sources: **Soapy BladeRF Source** and **Soapy PLUTO Source**, both with a Sample Rate of 1M and Center Freq (Hz) of 2.1G. Their outputs are combined in a **Virtual Source** (Stream ID: rx_signal_filtered). This signal enters a **Symbol Sync** block, which includes a Timing Error Detector (y[n]y'[n] Maximum likelihood), Samples per Symbol: 6, Expected TED Gain: 1, Loop Bandwidth: 45m, Damping Factor: 1, Maximum Deviation: 1.5, Output Samples/Symbol: 1, and an Interpolating Resampler (MMSE, 8 tap FIR). The output of Symbol Sync is split: one path goes to a **Costas Loop** (Loop Bandwidth: 62.8319m, Order: 4), and the other goes to a **QT GUI Time Sink** (Number of Points: 1.024k, Sample Rate: 1M, Autoscale: No). The Costas Loop outputs frequency, phase, and error signals. The error signal is sent to a **Constellation Decoder** (Constellation Object: ...f70>). The decoder's output goes to an **Unpack K Bits** block (K: 2, 0123 -> 01). The unpacked bits are then processed by a **Packed to Unpacked** block (Bits per Chunk: 2, Endianness: MSB) and sent to a **Virtual Sink** (Stream ID: ber_output_signal). This virtual sink is connected to two external sinks: **Soapy BladeRF Sink** and **Soapy PLUTO Sink**, both with a Sample Rate of 1M and Center Freq (Hz) of 2.1G. The output of the Packed to Unpacked block also goes to a **File Sink** (File: ber_output.txt, Unbuffered: Off, Append file: Append). The output of the Constellation Decoder also goes to a **File Sink** (File: ber_output.txt, Unbuffered: Off, Append file: Append). The output of the File Sink is sent to a **QT GUI Time Sink** (Name: Unpacked data, Number of Points: 1.024k, Sample Rate: 1M, Autoscale: No).

The BER calculation block diagram shows the flow from two **Virtual Sources** (Stream ID: ber_input_signal and Stream ID: ber_output_signal) to a **BER** block (Test Mode: False). The output of the BER block is sent to a **BER Align** block (Cal_Size: 40.96k, Report_Every: 1.024k). The output of the BER Align block is sent to a **QT GUI Number Sink** (ID: qtgui_number_sink_0, Autoscale: No, Average: 0, Graph Type: Horizontal).